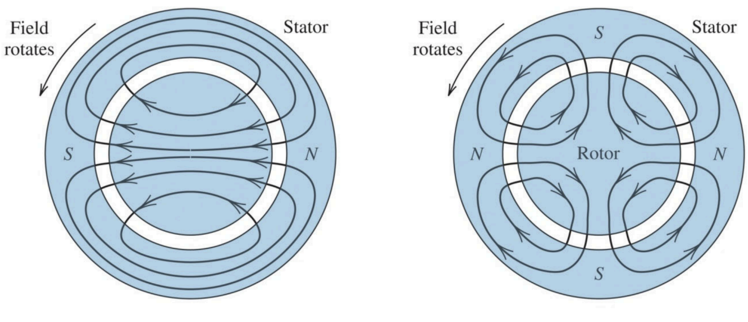
AC Motors

Three Phase Induction Motors

Working Principle

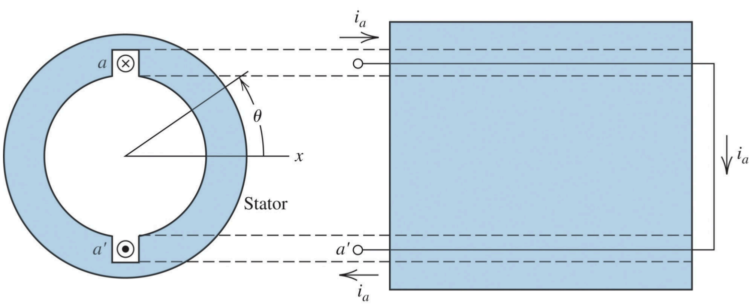
The most common type of AC motor is the three phase AC induction motor, typically producing power of at least 5hp.

Generally, the working principle of three phase induction motors is that a three phase AC power source connects to three windings in the stator of the motor. The windings are arranged in a way that a magnetic field is induced, which rotates as the AC current varies. The induced magnetic field will also has an even number of poles.



The windings in the rotor have an induced current due to the magnetic field, which cause a turning force to be induced in the rotor windings.

If we focus on one winding, and simplify it to be a single coil:



The flux produced by the winding is proportional to the cosine of the rotation of the winding, or in other words

$$B_a = Ki_a(t)\cos(\theta)$$

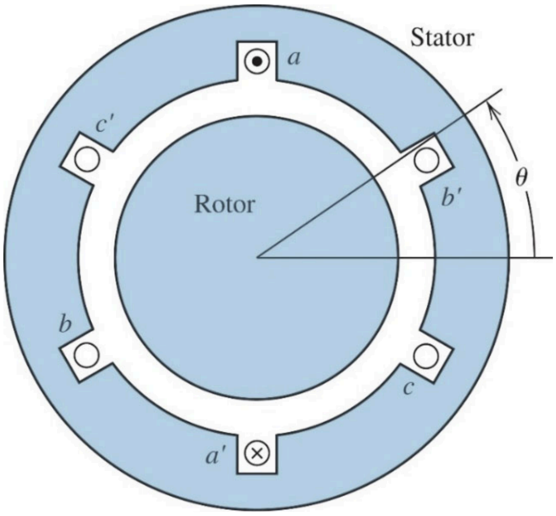
This also means that B is a maximum when theta is at 0 or 180.

If we let $i_a(t) = I_m \cos(\omega t)$,

$$B_a = KI_m \cos(\omega t)\cos(\theta)$$

This shows that the magnetic flux varies with time, and oscillates, while also varies with the fixed rotation in space.

The idea is that 2 other windings are added at 120 degrees apart, hence the magnetic flux



$$B_a = Ki_a(t)\cos(\theta)$$
$$B_b = Ki_b(t)\cos(\theta - 120^\circ)$$
$$B_c = Ki_c(t)\cos(\theta - 240^\circ)$$

Hence the total magnetic field is the sum of the three flux

$$\begin{aligned} B_{gap} &= B_a + B_b + B_c \\ &= Ki_a(t)\cos(\theta) + Ki_a(t)\cos(\theta - 120^\circ) + Ki_a(t)\cos(\theta - 240^\circ) \\ &= KI_m(\cos(\omega t)\cos(\theta) + \cos(\omega t - 120^\circ)\cos(\theta - 120^\circ) + \cos(\omega t - 240^\circ)\cos(\theta - 240^\circ)) \end{aligned}$$

which can be derived similarly to wye-wye power to get

$$B_{gap} = \frac{3}{2}KI_m\cos(\omega t - \theta)$$

Or in other words, the maximum magnetic flux is

$$B_m = \frac{3}{2}KI_m$$

Notice how this maximum flux is achieved when $\cos(\omega t - \theta) = 1$, or when

$$\theta = \omega t$$

This means that the maximum flux is achieved when the winding is positioned at ωt, but since this value is changing with time, this means that the flux produced is rotating, in the counter clockwise direction.

If $\theta = -\omega t$ for the flux to be maximum, the flux is rotating clockwise, which is done by changing the 3 phase input to a negative phase sequence.

Poles & Synchronous Speed

For a induction motor that induces P number of poles, the angular speed is

$$\omega_s = \omega \times \frac{2}{P}$$

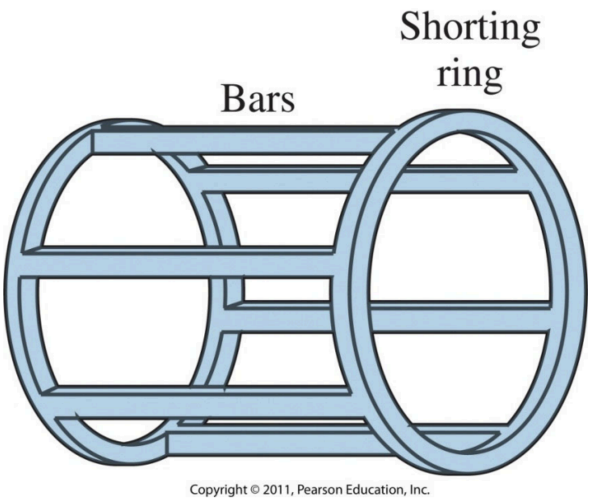
where ω_s is the synchronous speed and ω is the AC angular frequency. The synchronous speed can be found as a function of the current's properties, that is

$$\omega_s = \frac{2\pi}{60} \times n_s = \frac{2\pi}{60} \times \frac{120f}{P}$$

Notice how the synchronous speed is inversely related to the number of poles.

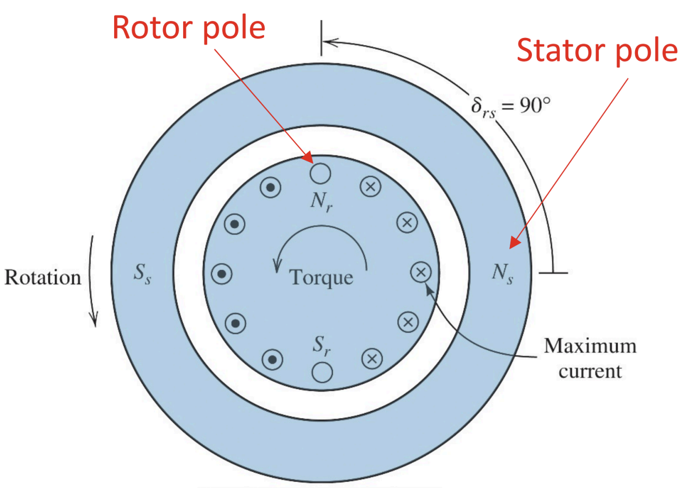
Squirrel Cage Induction Motor

The squirrel cage induction motor is a type of inductor motor where there are multiple rotor windings in the form of a "squirrel cage":



The squirrel cage consists of aluminium bars with a shorting ring at each end. Typically, the squirrel cage is formed by casting aluminium into an iron core with slots.

The squirrel cage is not supplied by any external power source, and is induced by the rotating magnetic field in the stator. This also gives it an advantage to DC motors, as bushes and commutators are not needed and hence do not rub more on the rotor, allowing for longer motor life.



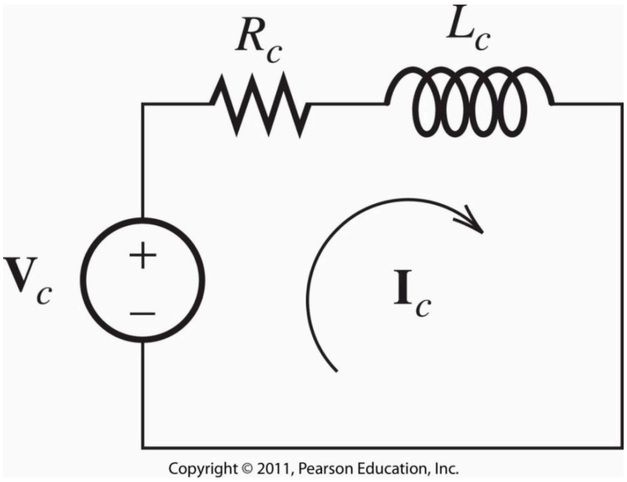
The working principle of the squirrel cage motor is that as the magnetic flux produced by the stator rotates, a voltage is induced along each bar, in the form of

$$V_c = Blu$$

where u is the relative speed between the conductor and the rotating field.

Due to the shorting rings, current flows through the bars, which causes the whole cage to have its own poles set up, 90 degrees to the stator's poles. This causes a turning effect as the rotor wants to align its poles to the stator, but due to the rotating magnetic field, it continues to rotate continuously.

Shown is the equivalent circuit of the conductor winding. Due to the conductor being embedded in iron, there is a non negligible amount of inductance, hence we represent the conductor as a series resistance and inductance.



If we know that the frequency of the induced frequency is ω_s, we can find the impedance as

$$Z = R_c + j\omega_{slip}L_c$$

And hence the current is

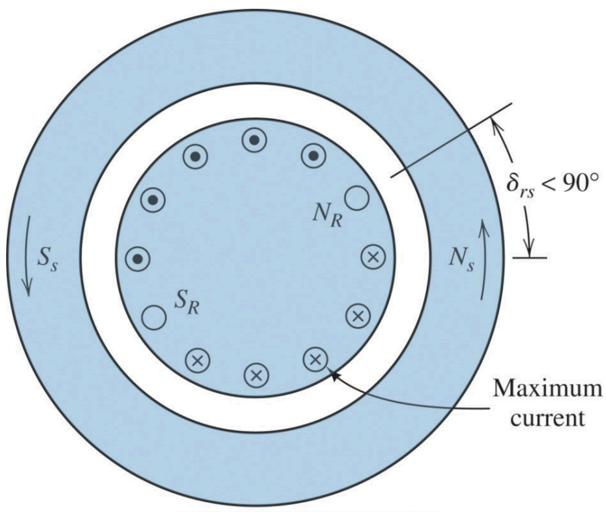
$$\mathbf{I}_c = \frac{\mathbf{V}_c}{R_c + j\omega_{slip}L_c}$$

Since the inductance is in the form of $Z\angle\theta$, where $0° \leq \theta \leq 90°$,

$$\mathbf{I}_c = \frac{\mathbf{V}_c}{Z\angle\theta} = \frac{V_m\angle\phi}{Z\angle\theta} = \frac{V_m}{Z}\angle\phi - \theta$$

which means that current lags the voltage. This lag causes the angle between the poles to not be fully 90 degrees:

Rotor pole is displaced by δ_{rs} < 90°



Due to the non perpendicular poles, a reduction of torque is caused

Slip & Slip Frequency

Recall that we let the induced frequency be ω_{slip}. This frequency is dependent on the relative speed between the stator and the motor, and number of poles P.

Recall the synchronous speed ω_s, which is actually the same as the speed the stator winding's field rotates at. If we know the motor's speed ω_m, which is the same speed the rotor flux rotates at due to the fixed connection, the speed of the stator relative to the rotor is

$$\omega_s - \omega_m$$

We define slip (or more specifically normalised slip) as the ratio of the relative speed to the stator speed, as

$$s = \frac{\omega_s - \omega_m}{\omega_s} = \frac{n_s - n_m}{n_s}$$

which is a factor from 0 to 1. Slip represents the loss in frequency from the supplied AC power to the induced AC current in the rotor.

Hence, this influences the slip frequency by

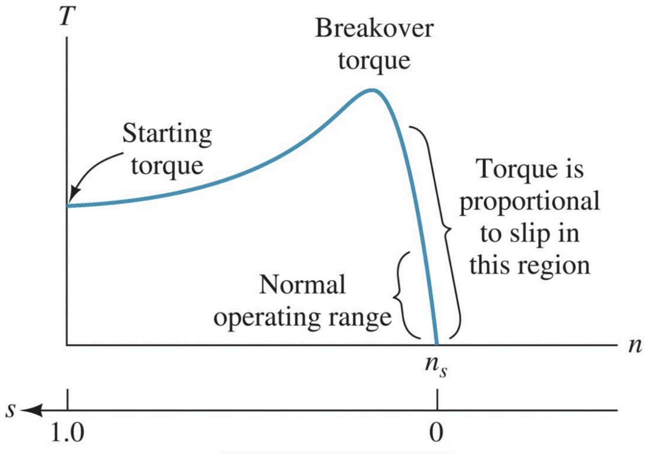
$$\omega_{slip} = s\omega \Rightarrow f_{slip} = sf$$

where ω is the AC angular frequency.

Notice how the frequency of induced voltage is the same as the AC frequency when the motor speed is 0, or when the rotor is stationary, and the induced voltage has zero frequency when the rotor rotates at the same speed as the synchronous speed.

Torque-Speed Characteristics

Recall the torque-speed characteristic graph from DC motors:



We know that torque is equal to 0 at a certain motor speed n_s, which in terms of slip, the relative speed is 0 and the slip is 0. Hence, the induced voltage frequency is 0, which causes the rotor current to 0. Since the current causes the induced force, which causes the torque, the torque is 0 too.

For small slip values, the current is low and hence the conductor impedance can be assumed to be purely real (reactance is negligible), current and voltage are in phase and hence the poles are perfectly perpendicular to each other, which produces the optimal torque.

We know that slip is directly proportional to the relative speed $\omega_s - \omega_m$, and that the induced voltage is directly proportional to the relative speed, by

$$V_c = Blu = Blr(\omega_s - \omega_m)$$

The two equations can be rearranged to show that the induced voltage is proportional to the slip, s.

Since we are still assuming the inductance is negligible, current is directly proportional to the voltage, hence the current is proportional to the slip. We know that the current causes the turning force in a proportional manner, thus we can conclude that at small values of s, torque is directly proportional to slip.

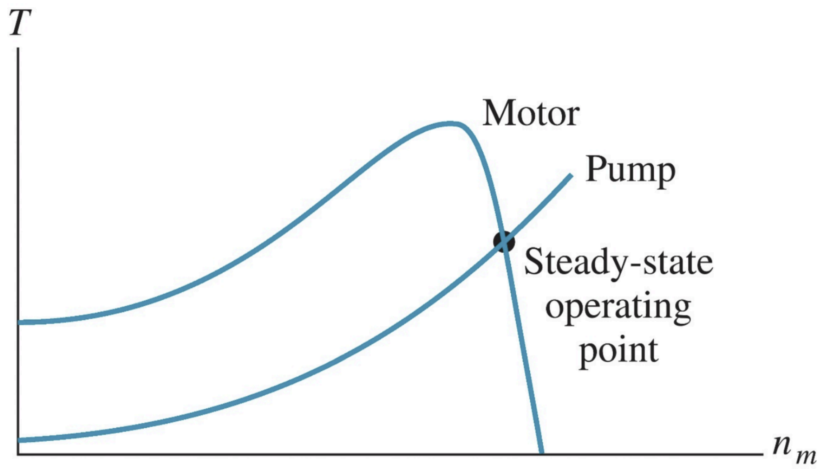
$$T \propto s$$

As we continue to slow the motor speed, slip increases and the torque increases, hence the current increases. This causes a significant amount of inductance to be present, which causes current to be nearly independent of slip (??), As current becomes constant, torque starts to level out, until it reaches the starting torque when the motor speed is 0.

This slowing of the motor and increased torque is due to the poles slowly aligning, and the poles fully align when the motor comes to a stop.

Other than the starting torque, another torque value to take note of is the breakdown torque; the maximum possible torque possible.

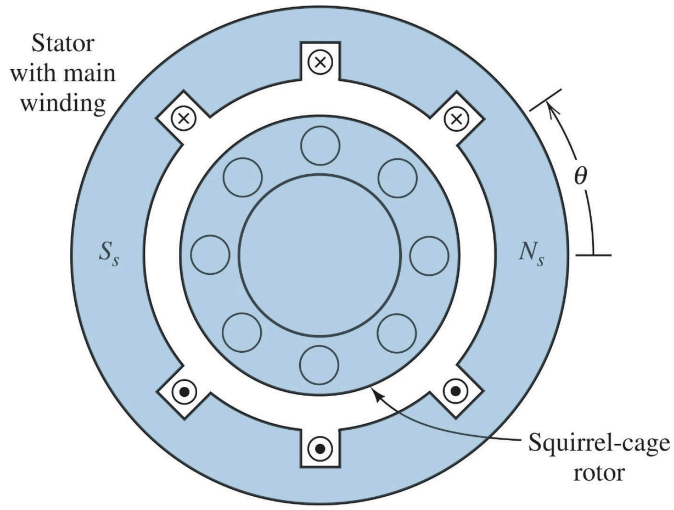
In the case where an inductor motor is connected to a load, say a pump:



The steady state of the motor will be when the torque produced by the motor is equal to the torque required by the load, or the intersection of the two characteristic curves.

Single Phase Motors

Single phase motors only require one ac power source as input. Shown is the sectional view of a single phase motor:



The motor still has a squirrel cage rotor, however there is only one main winding, but requires another winding called an auxiliary winding to start up the motor.

We know that the magnetic flux in the gap is

$$B = Ki(t)\cos(\theta)$$

and since only one phase of AC is present,

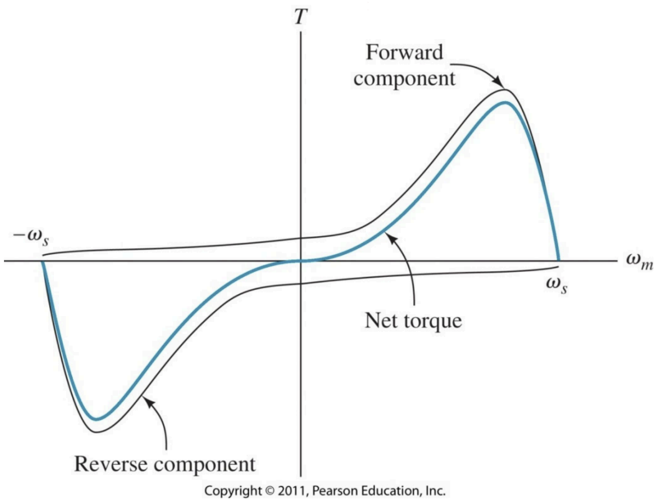
$$B = KI_m\cos(\omega t)\cos(\theta)$$

Since no other phase is present, this flux flips, or pulsates, switching direction twice for each cycle. To show this, rewrite B as

$$B = \frac{KI_m}{2}\left(\cos(\omega t - \theta) + \cos(\omega t + \theta)\right)$$

which shows two cosine terms that counter-rotate each other, showing that the magnetic flux flips. This pulsating magnetic flux induces voltage in the rotor and hence causes a torque and a rotation counter clockwise.

Shown is the torque-speed characteristic graph of a single phase AC motor:



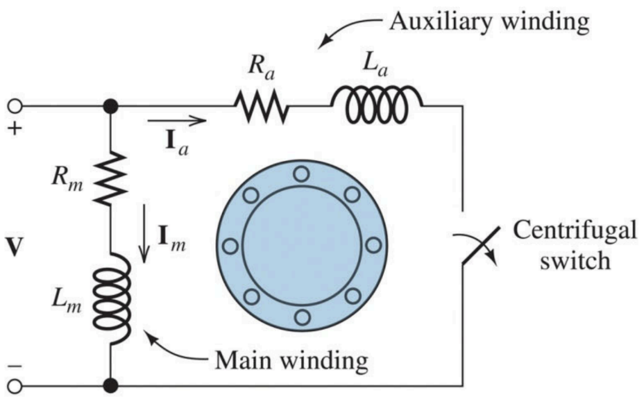
This is formed by the sum of the forward and negative component, similar to the two cosine terms in the magnetic flux found earlier.

Notice the similarity of it to the induction motor especially near the synchronous speed, however notice there is no starting torque, meaning that it cannot start a load from a standing start.

The graph is also symmetric, meaning that if the motor was running in the other direction, the operation will still be the same, hence it is capable of running in either direction.

Auxiliary Windings

Auxiliary windings seek to fix the issue of the lack of a starting torque. Typically, the auxiliary winding is placed 90 degrees to the main winding, and has a different impedance to the main winding to have the currents flowing through each winding be 90 deg apart from each other.



To achieve the different impedance, the auxiliary winding is wound more densely to change the inductance and hence the phase angle. This causes the auxiliary winding current to have a different phase angle than the main winding current. This induces a magnetic field and magnet poles that act as a small push to start the motor, and encourage the rotor to continue turning as it starts up.

After a certain speed, a centrifugal switch breaks and the main winding continues to keep the rotor's rotation.

Some common failures of this method is when the centrifugal switch does not open, causing the auxiliary winding to burn out as it is not designed to run the motor for a long time.

There is also some vibration between the switch from the rotor being induced by both windings, to being induced by one motor, which can be seen on its torque-speed characteristic graph as a sudden drop: